CHAPTER ONE

1.1 Background of the Study

The management of organic waste remains a major environmental concern, particularly in regions with increasing agricultural activities and limited waste treatment infrastructure. Among the various types of organic waste, cow dung is one of the most commonly generated and utilized materials, especially in rural and peri-urban communities. It is often applied directly to soil as an organic fertilizer due to its nutrient content and accessibility. However, the use of untreated cow dung presents potential environmental and public health risks, as it may contain high microbial loads, including pathogenic organisms, as well as substances capable of contaminating soil and water systems (Ndubuisi-Nnaji *et al*., 2022).

In response to these challenges, anaerobic digestion has gained attention as a sustainable method for managing organic waste. This biological process converts organic materials into biogas, a renewable energy source, while producing digestate as a residual by-product. The process is widely regarded as an effective means of reducing waste volume and stabilizing organic matter, thereby improving its potential for reuse (Álvaro *et al*., 2024).

Digestate is commonly considered a safer alternative to raw organic waste and is frequently applied as a soil amendment due to its nutrient-rich composition, which supports soil fertility and agricultural productivity (Decorte *et al,* 2024) . However, its characteristics and safety are largely influenced by the type of feedstock used and the operational conditions of the digestion process (Ndubuisi-Nnaji *et al*., 2022).

Despite these perceived benefits, concerns have been raised regarding the environmental biosafety of digestate. Residual microorganisms, toxic compounds, and other potentially harmful substances may persist after digestion, posing risks when introduced continuously into the environment. In addition, the reduction of harmful components during anaerobic digestion is not always complete, and variations in process efficiency may affect the final quality and safety of the digestate (Ndubuisi-Nnaji *et al*., 2022).

Given these concerns, there is a need to critically assess and compare the biosafety risks associated with untreated cow dung and digestate from a batch biogas reactor. Such evaluation is essential for determining their suitability for environmental and agricultural applications and for promoting safer and more sustainable waste management practices.

1.2 Statement of the Problem

The widespread use of untreated cow dung as an organic fertilizer, particularly in areas with limited waste treatment practices, presents potential environmental and public health risks.

Despite its agricultural benefits, untreated cow dung may introduce harmful microorganisms and other contaminants into soil and water systems, raising concerns about its safety for environmental application.

Anaerobic digestion has been adopted as an alternative method for managing organic waste, with the expectation that it reduces these risks while producing digestate as a by-product. Digestate is often regarded as a safer material and is commonly applied to land without thorough evaluation of its biosafety status. However, this assumption may not always be valid, as the effectiveness of anaerobic digestion in eliminating harmful components depends on several factors, including process conditions and feedstock characteristics.

The main problem lies in the lack of clear and consistent information on the relative safety of untreated cow dung and digestate. In many cases, both materials are used in agricultural without sufficient assessment of their potential risks. This creates uncertainty regarding their impact on soil quality, water resources, and public health.

Furthermore, there is limited comparative analysis that clearly determines whether anaerobic digestion significantly improves the safety of organic waste materials or if certain risks persist after treatment. Without such evaluation, it is difficult to make informed decisions on the safe use and management of these materials.

Therefore, there is a need for a systematic assessment of the environmental biosafety risks associated with untreated cow dung and digestate from a batch biogas reactor in order to determine their relative safety and support appropriate environmental management practices

.

1.3 Research Questions

The following research questions were formulated to guide this study:

1. What are the risks associated with untreated cow dung?
2. What are the risks associated with digestate from a batch biogas reactor?
3. Does anaerobic digestion reduce the risks associated with organic waste?
4. Are untreated cow dung and digestate safe for environmental or agricultural use?
5. Which is safer, untreated cow dung or digestate?

1.4 Aim and Objectives

Aim of the Study

The aim of this study is to assess the environmental biosafety risks associated with untreated cow dung and digestate from a batch biogas reactor.

Objectives of the Study

The specific objectives of the study are to:

1. Determine the the total heterotrophic bacterial count present in untreated cow dung and digestate.
2. Evaluate the anaerobic bacterial count in untreated cow dung and digestate.
3. Determine the total coliform count in both untreated poultry manure and digestate samples.
4. Assess the fecal coliform (E.Coli) count as an indicator of fecal contaminationin the samples.
5. Enumerate the salmonella and shigella species present in untreated poultry manure.

1.5 Research Hypothesis

Null Hypothesis (H_0): There is no significant difference in the environmental biosafety risks between untreated cow dung and digestate from a batch biogas reactor.

Alternative Hypothesis (H_1): There is a significant difference in the environmental biosafety risks between untreated cow dung and digestate from a batch biogas reactor.

1.6 Significance of the Study

This study is significant as it contributes to a better understanding of the environmental and public health implications associated with the use of untreated cow dung and digestate from biogas systems. By evaluating and comparing their biosafety risks, the research provides valuable insight into the potential hazards these materials may pose when introduced into the environment.

The findings of this study will be beneficial to environmental managers and regulatory bodies by providing scientific evidence that can guide policies and standards for the safe handling, treatment, and application of organic waste materials. It will also assist in improving waste management practices, particularly in regions where untreated livestock waste is commonly applied to agricultural land without proper assessment.

In addition, this research will be useful to farmers and practitioners involved in the use of organic fertilizers, as it will provide information on the safety and suitability of both untreated cow dung and digestate for agricultural purposes. This can help in minimizing risks associated with soil contamination, water pollution, and the spread of harmful microorganisms.

Furthermore, the study contributes to the field of environmental management and toxicology by expanding existing knowledge on biosafety risk assessment of cow dung. It will serve as a reference for future research, particularly in areas focusing on waste-to-energy technologies and their environmental implications.

Overall, the study promotes safer and more sustainable approaches to organic waste utilization, thereby supporting environmental protection and public health.

1.7 Scope of the Study

This study focuses on the environmental biosafety risk assessment of untreated cow dung and digestate obtained from a batch biogas reactor. It is limited to the evaluation of selected biosafety parameters that are relevant to environmental and public health, including microbiological and physicochemical characteristics.

The study involves the collection and analysis of untreated cow dung samples as well as digestate samples produced from a controlled batch anaerobic digestion process. Emphasis is placed on assessing the presence and levels of potential microbial contaminants and other indicators that may pose risks to soil, water, and human health.

Furthermore the findings are based on controlled experimental conditions and may not fully represent variations that could occur under different environmental or operational settings.

In addition, the study does not cover the economic evaluation of biogas production or the engineering design of biogas systems, but rather concentrates on the biosafety implications of the materials involved.

1.8 Limitations of the Study

This study is subject to certain limitations that may influence the scope and interpretation of the results. One of the major limitations is the challenge of maintaining a constant power supply, which is necessary for sustaining the optimal temperature required for the anaerobic digestion process. Fluctuations in electricity supply may affect the efficiency of the biogas reactor and, consequently, the characteristics of the digestate produced.

Time constraints also limit the duration of the study, making it difficult to observe long-term variations in the biosafety properties of the samples. In addition, limited laboratory resources and equipment may restrict the number of parameters that can be analyzed, thereby narrowing the scope of the assessment.

Furthermore, the study is based on samples obtained from specific sources under controlled experimental conditions. As such, the findings may not fully represent variations that could occur under different environmental conditions or with different types of feedstock.

Despite these limitations, appropriate procedures are followed to ensure that the results obtained are reliable and relevant within the scope of the study.

1.9 Operational Definition of Terms

Cow Dung: The fecal matter excreted by cattle, used in this study in its untreated form for biosafety assessment.

Digestate: The semi-solid or liquid residue produced after the anaerobic digestion of organic materials in a batch biogas reactor.

Anaerobic Digestion: A biological process in which microorganisms break down organic matter in the absence of oxygen, resulting in the production of biogas and digestate.

Batch Biogas Reactor: A type of anaerobic digester in which organic materials are loaded at once and allowed to undergo digestion for a fixed period without continuous input or output.

Biosafety: The measures and assessment procedures used to evaluate and control risks posed by biological materials to human health and the environment.

Environmental Biosafety Risk: The potential of a substance to cause harm to the environment, including soil, water, ecosystems, and human health, due to its biological or chemical components.

Microbial Load: The quantity or concentration of microorganisms present in a sample, used as an indicator of contamination or potential health risk.

Physicochemical Properties: The physical and chemical characteristics of a substance, including parameters such as pH, temperature, nutrient content, and chemical composition.

Pathogens: Microorganisms, such as bacteria, viruses, or parasites, that are capable of causing disease in humans, animals, or plants.